Data Collection & EDA

for Eye Blink Detection Dataset

Webcam-based eye-state and blink detection dataset



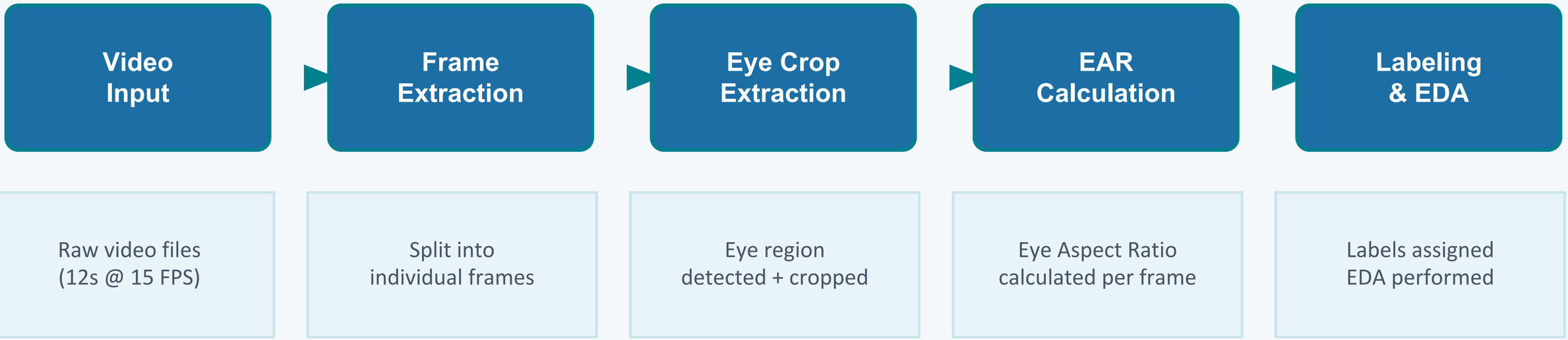
Dataset Collection Protocol

Collection Setup

- Laptop webcam — standard hardware
- Resolution: 640 × 480 pixels
- Frame rate: 15 FPS
- Duration: 12 seconds per video
- Varied lighting conditions
- Multiple camera distances
- With and without glasses
- Different behavioral scenarios

ID	Member Name	MemberID	ScenarioID	EnvID	GlassID	DistID
1	Hoàng	M01	SC1	E01	G0	D1
2	Hoàng	M01	SC1SQ	E02	G0	D1
3	Hoàng	M01	SC2	E03	G0	D1
4	Hoàng	M01	SC5	E04	G0	D1
5	Bảo	M02	SC1LD	E02	G0	D1
6	Bảo	M02	SC3	E04	G0	D1
7	Bảo	M02	SC4	E01	G0	D0
8	Bảo	M02	SC5	E03	G0	D2
9	Hiền	M03	SC1	E02	G1	D1
10	Hiền	M03	SC1SQ	E04	G1	D1
11	Hiền	M03	SC2	E01	G1	D0
12	Hiền	M03	SC3	E03	G1	D2
13	Trân	M04	SC1LD	E03	G0	D0
14	Trân	M04	SC4	E04	G0	D1
15	Trân	M04	SC5	E02	G0	D1
16	Trân	M04	SC1	E01	G0	D2
17	Nguyên	M05	SC1SQ	E03	G1	D2
18	Nguyên	M05	SC2	E04	G1	D1
19	Nguyên	M05	SC3	E02	G1	D0
20	Nguyên	M05	SC4	E01	G1	D1
21	Kiệt	M06	SC1LD	E01	G0	D1
22	Kiệt	M06	SC5	E04	G0	D0
23	Kiệt	M06	SC2	E02	G0	D2
24	Kiệt	M06	SC3	E03	G0	D1

Processing Pipeline



Pipeline Summary

- Raw videos are split into frames, then eye regions are extracted
- EAR is computed per frame; labels are assigned; data is cleaned for EDA

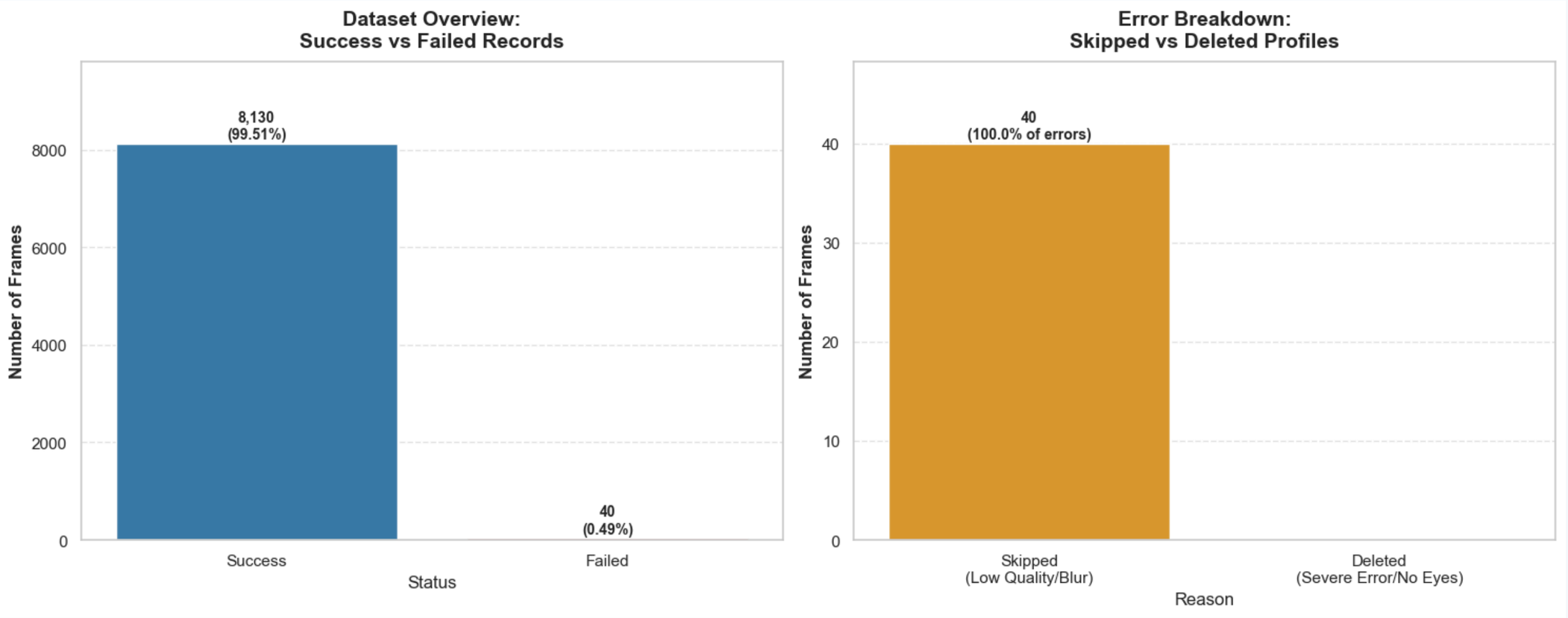
Dataset Scale & Integrity Audit

8,170
Total metadata records

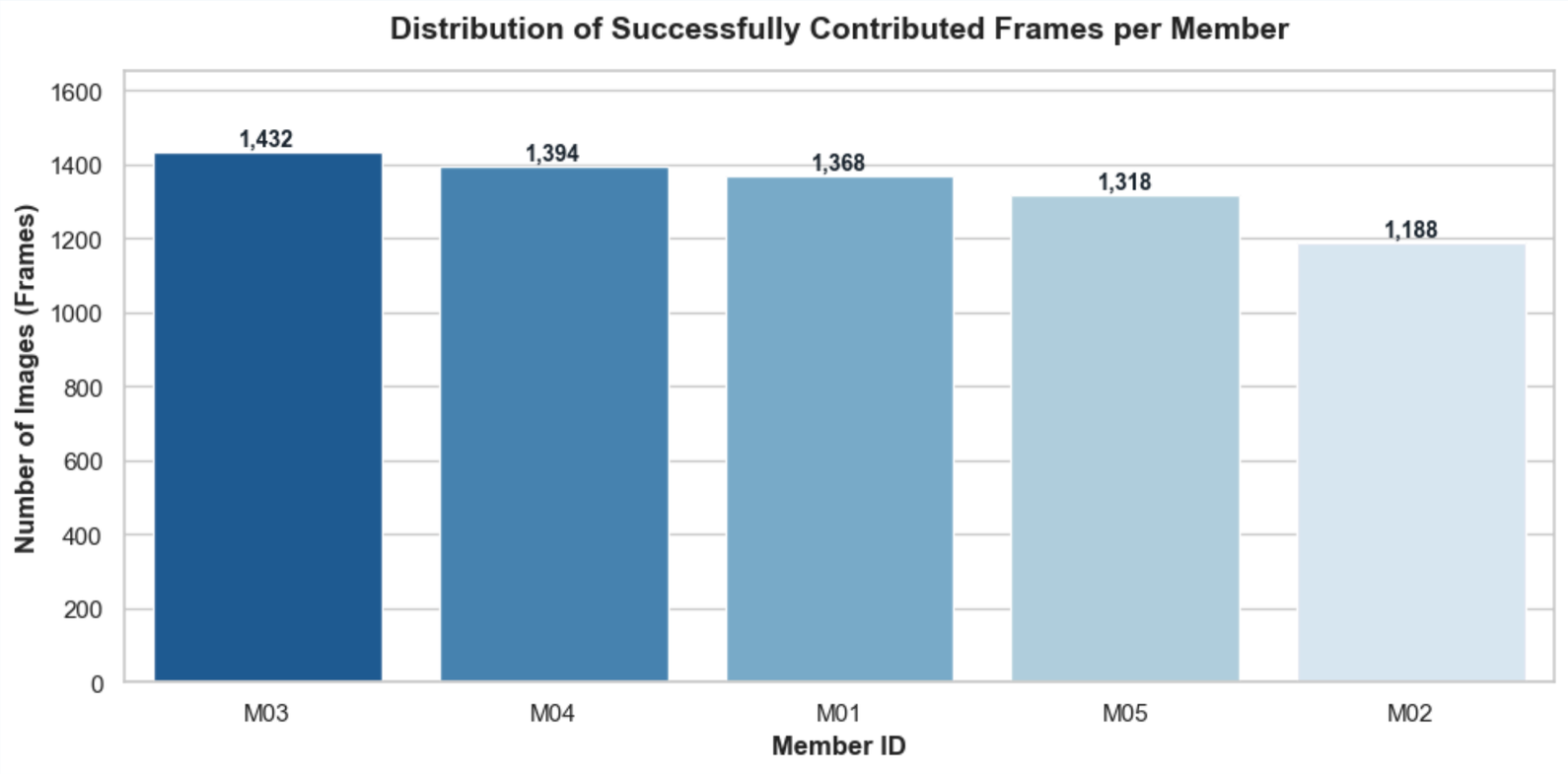
8,130
Successful samples

40
Failed / missing samples

0.49%
Data loss rate



Member Contribution Distribution



Key Observations

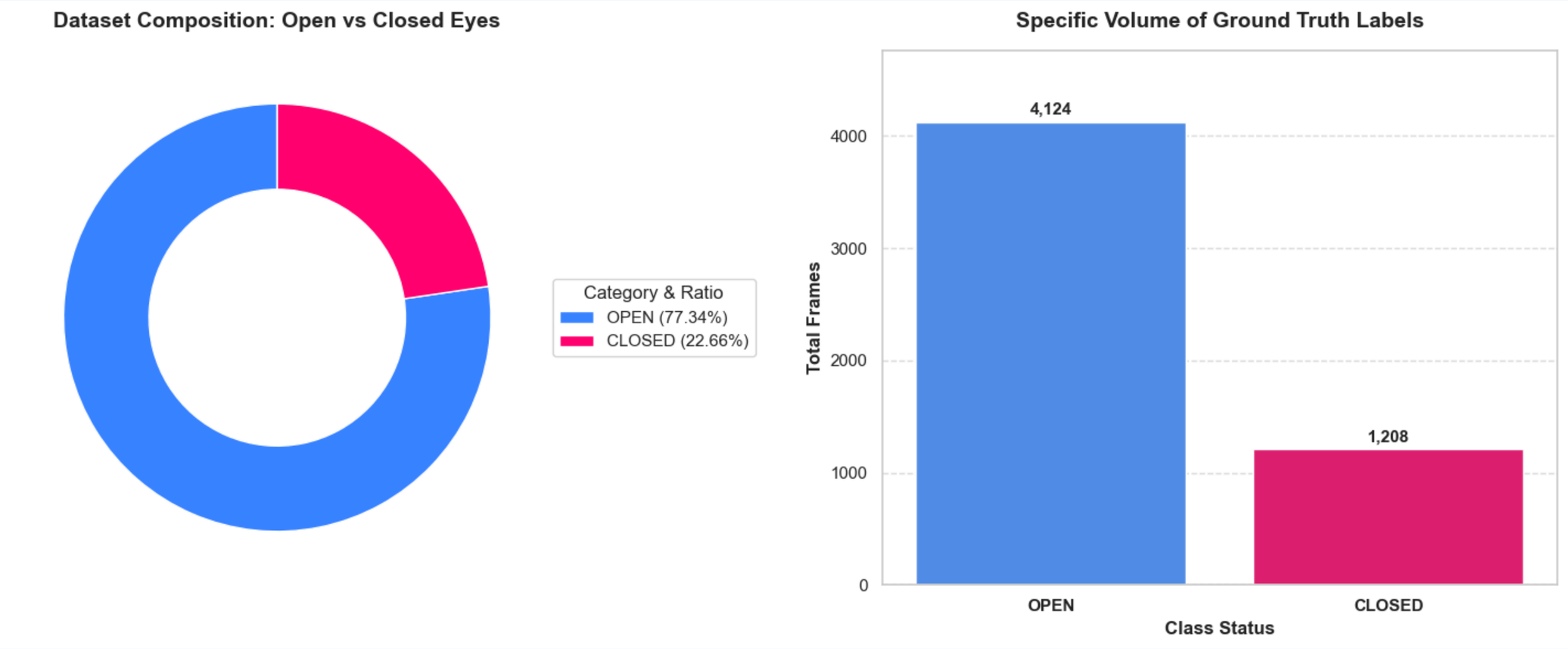
- Contributions are relatively balanced
- M03 contributes the most: 1,432 frames
- Lowest contribution: M02 with 1,188 frames
- No single contributor dominates the dataset
- Subject-aware evaluation is still needed

Class Balance Analysis

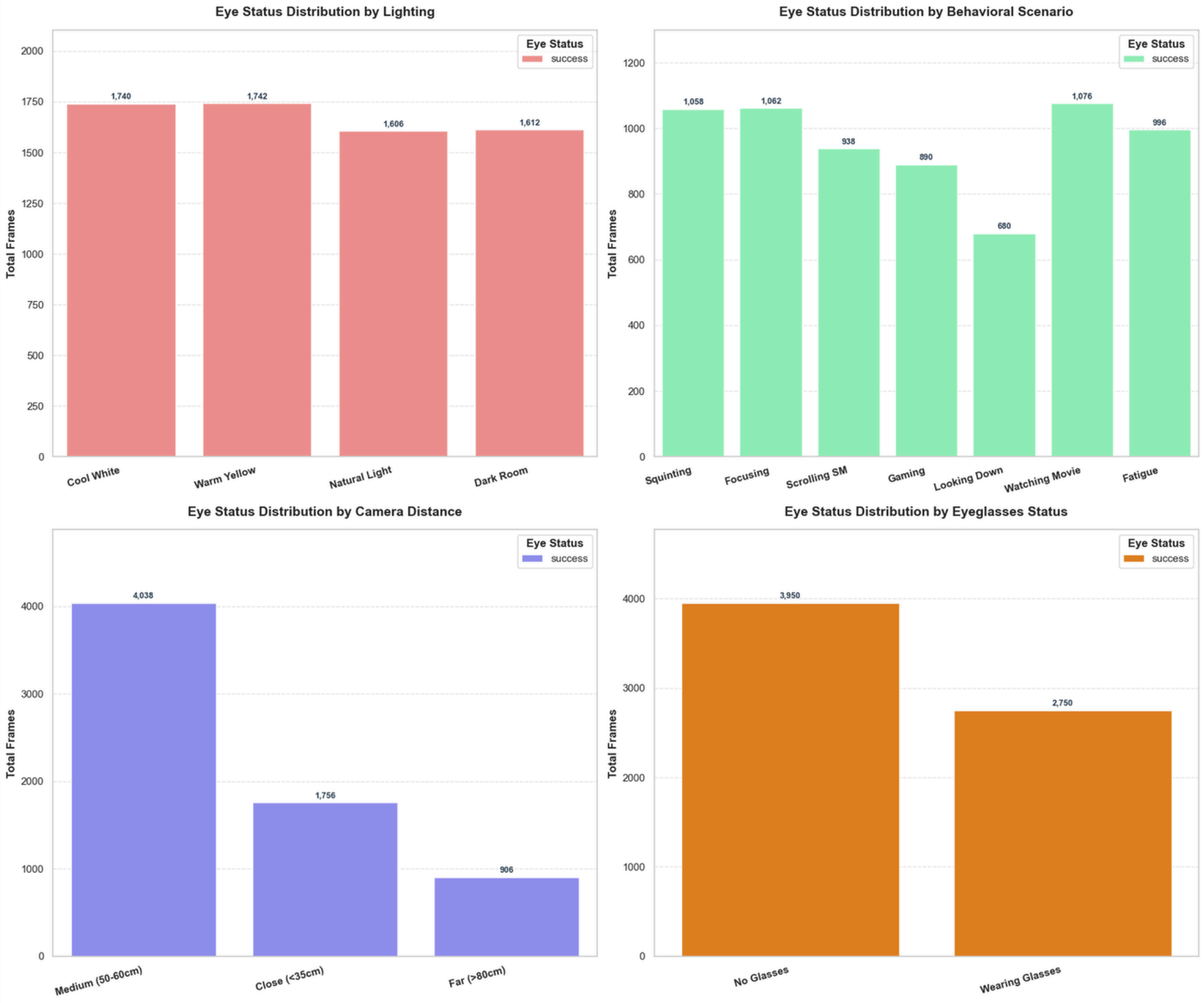
4,124
Open eye frames (77.34%)

1,208
Closed eye frames (22.66%)

⚠ **Class imbalance must be addressed**



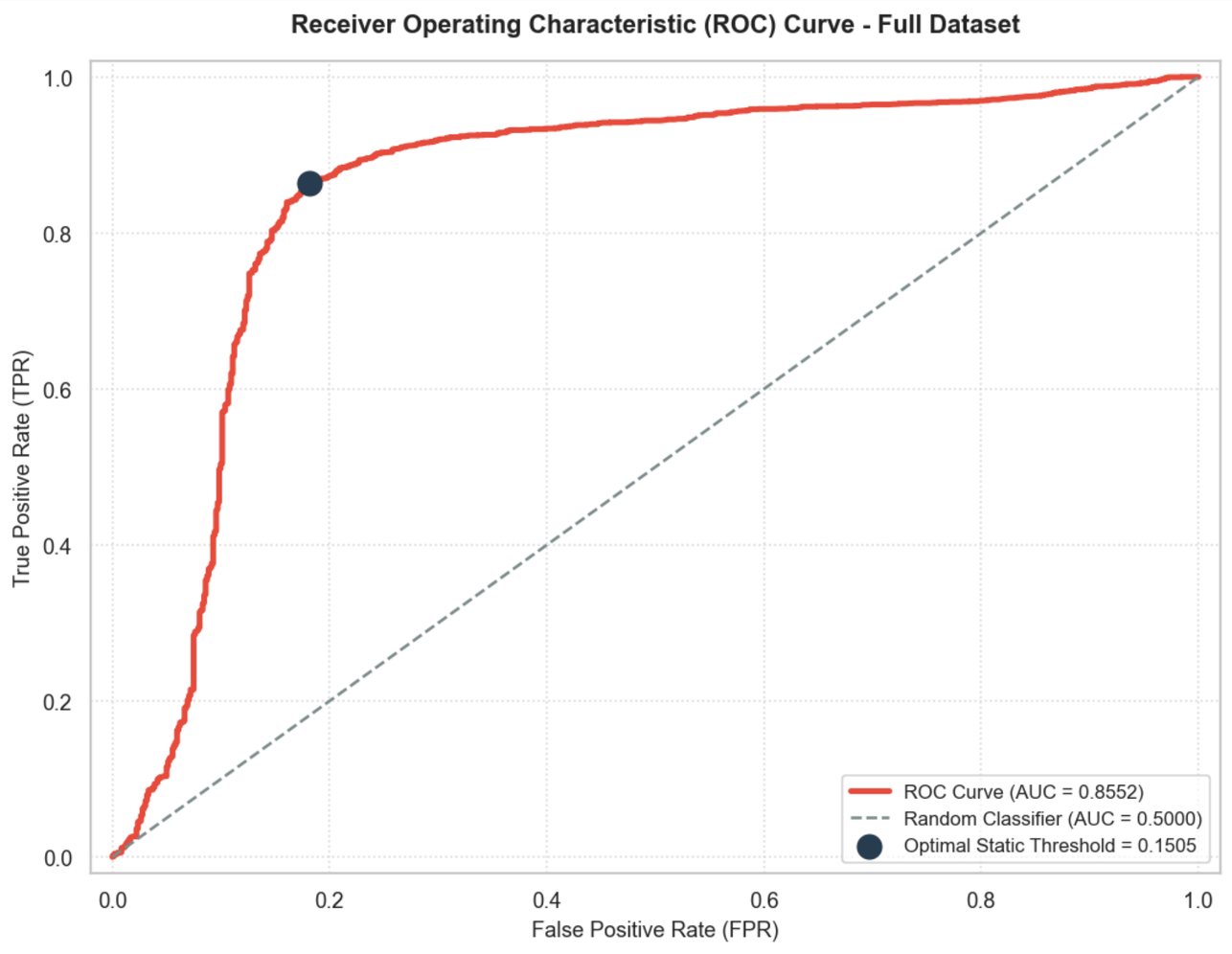
Experimental Context Coverage



Key Observations

- Lighting conditions are relatively well covered.
- Behavioral scenarios include multiple real-world cases.
- Medium camera distance has the highest number of samples.
- No-glasses samples are more common than glasses samples.
- Condition-aware evaluation is needed for robust testing.

Static EAR Threshold Baseline



The static threshold should be treated as a baseline, not the final solution.

Baseline Metrics

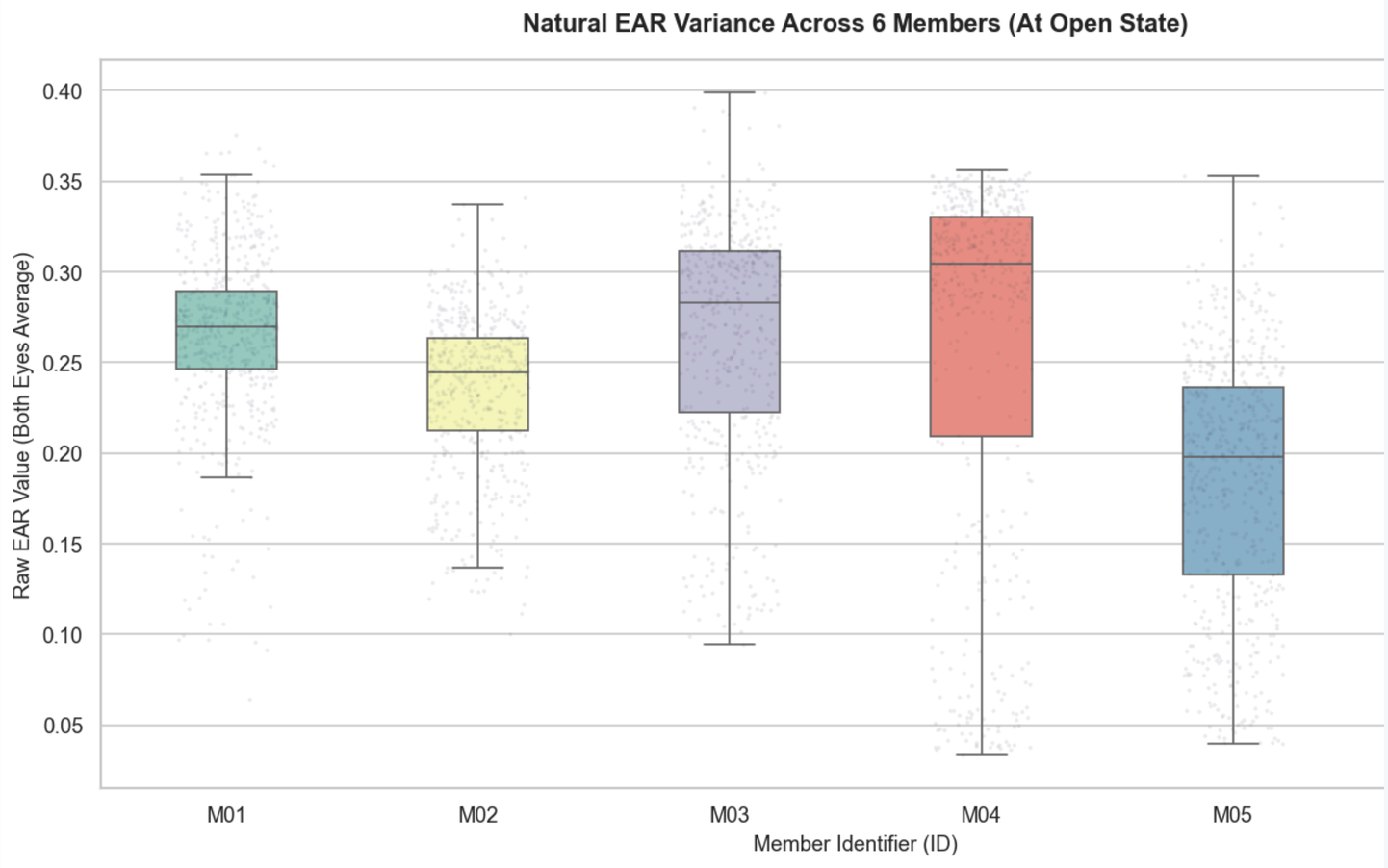
0.8552
ROC-AUC Score

0.1505
Optimal EAR Threshold

86.34%
True Positive Rate

18.14%
False Positive Rate

EAR Variance Across Members

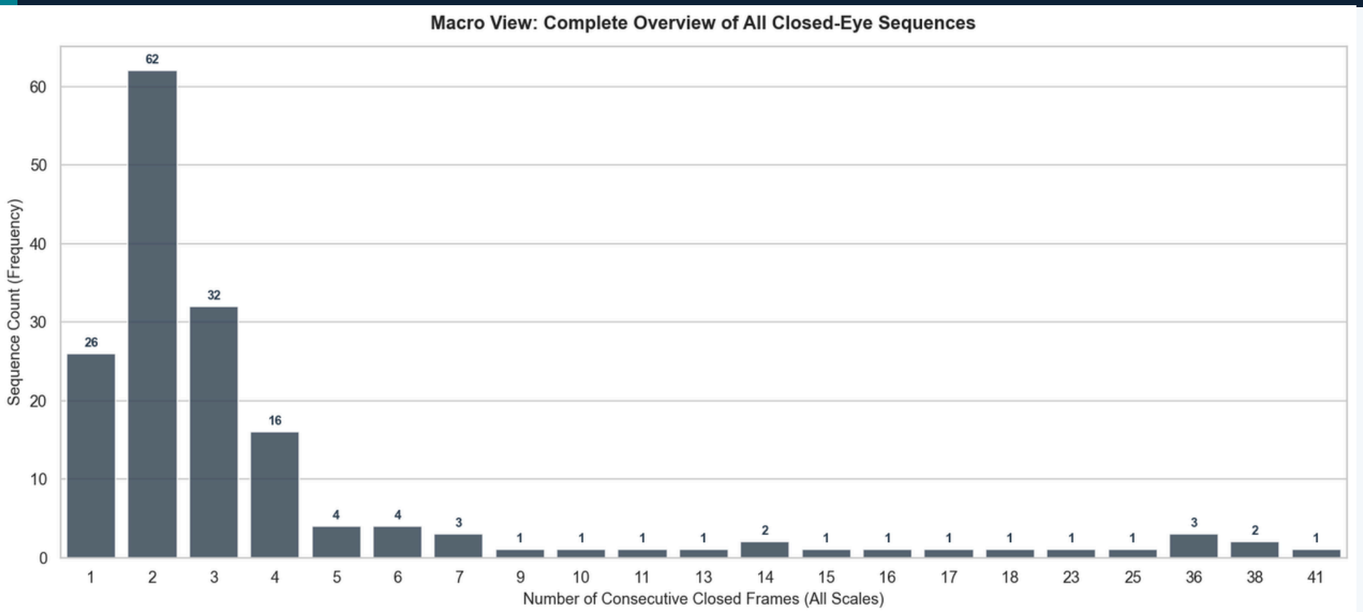


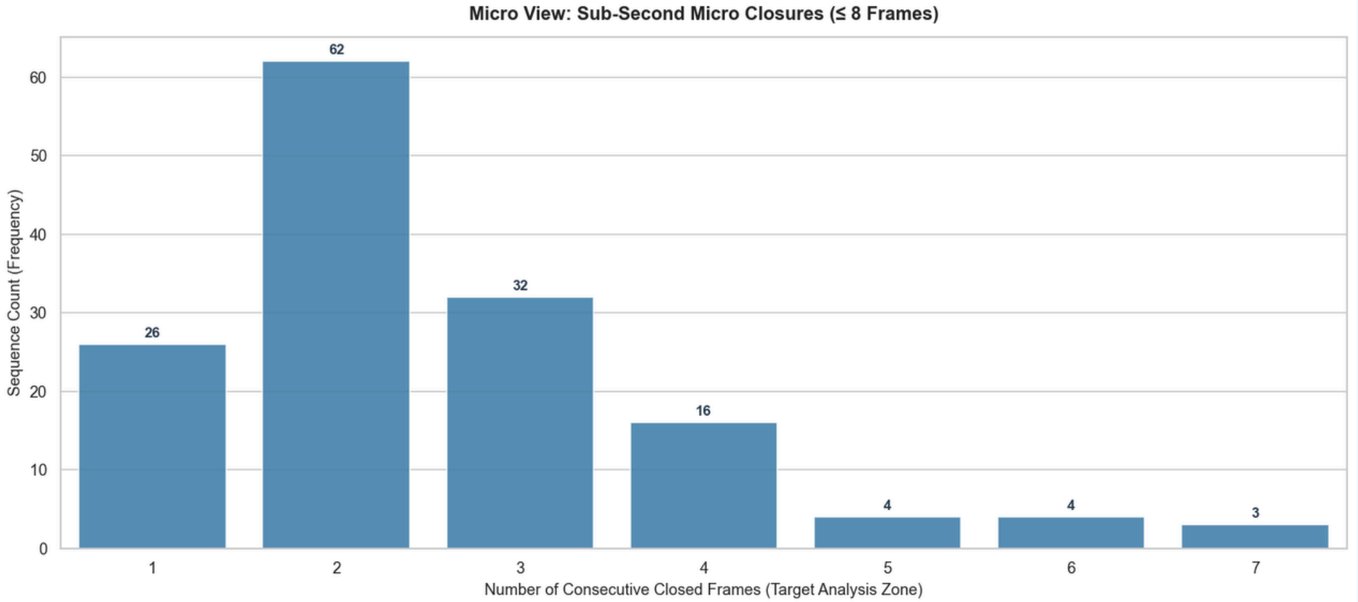
Implications

- The data of our members at the open-eye state
- Caused by personal eye shape & pose
- Camera distance also affects EAR range
- Static threshold may not fit all subjects
- Subject-aware normalization recommended

A single EAR threshold is unlikely to generalize across all subjects.

Closed-Eye Sequence Distribution





- Data includes intentional eye-closing for label balance.
- Actual blink duration falls between 1 to 7 frames.

Macro Overview

1
Min frames

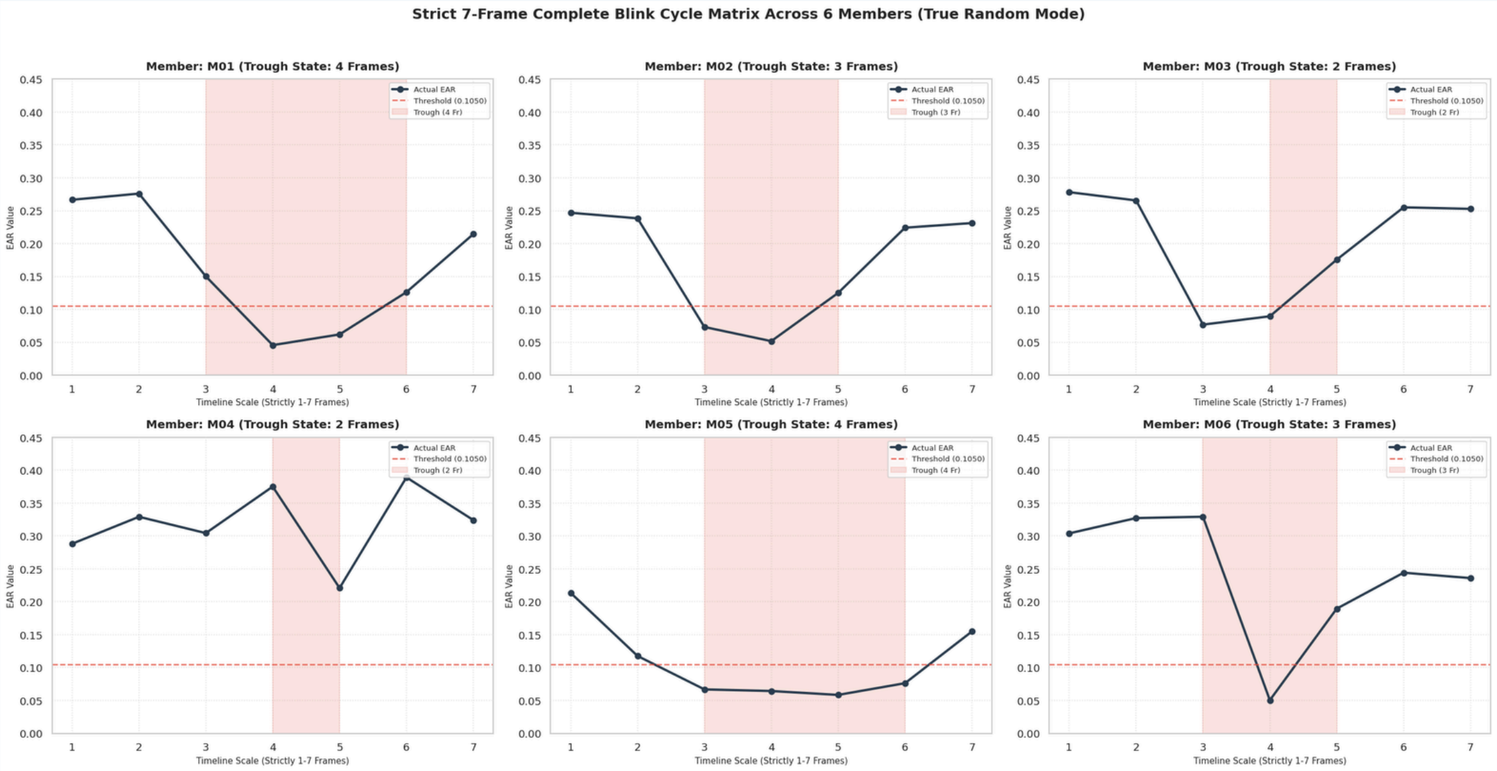
7
Max frames

Micro View (≤ 7 Frames)

2
Median frames

2.55
Mean frames

Sliding Window Decision



Recommendation

7 frames

Recommended window size

0.47 sec

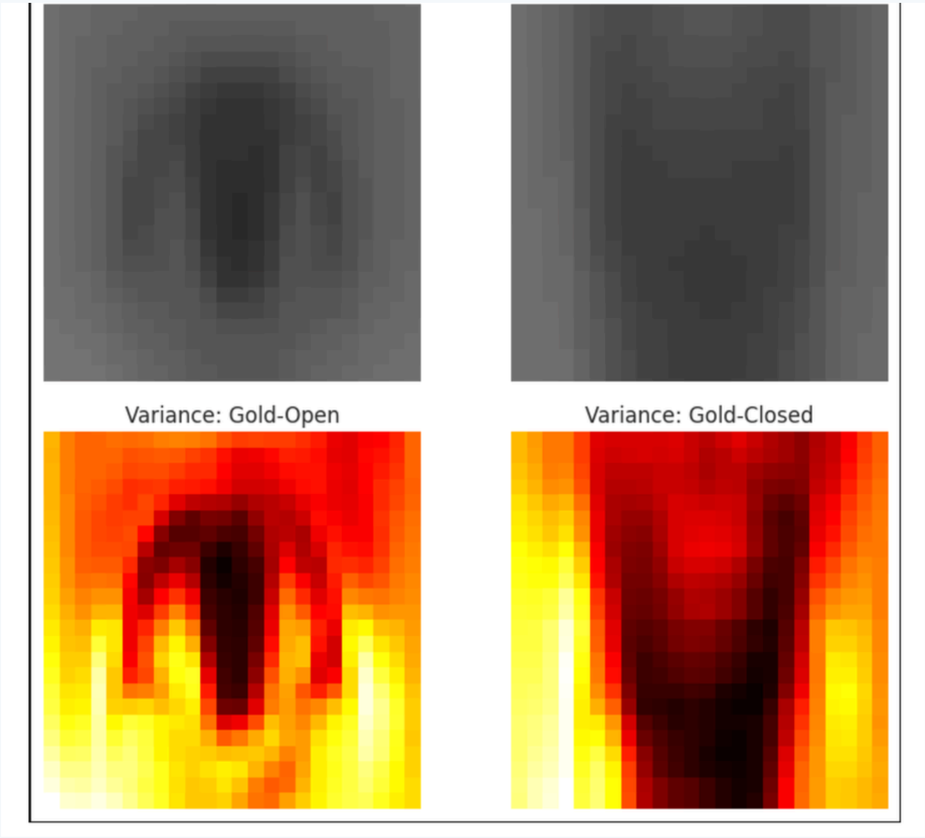
Duration at 15 FPS

Temporal Pattern

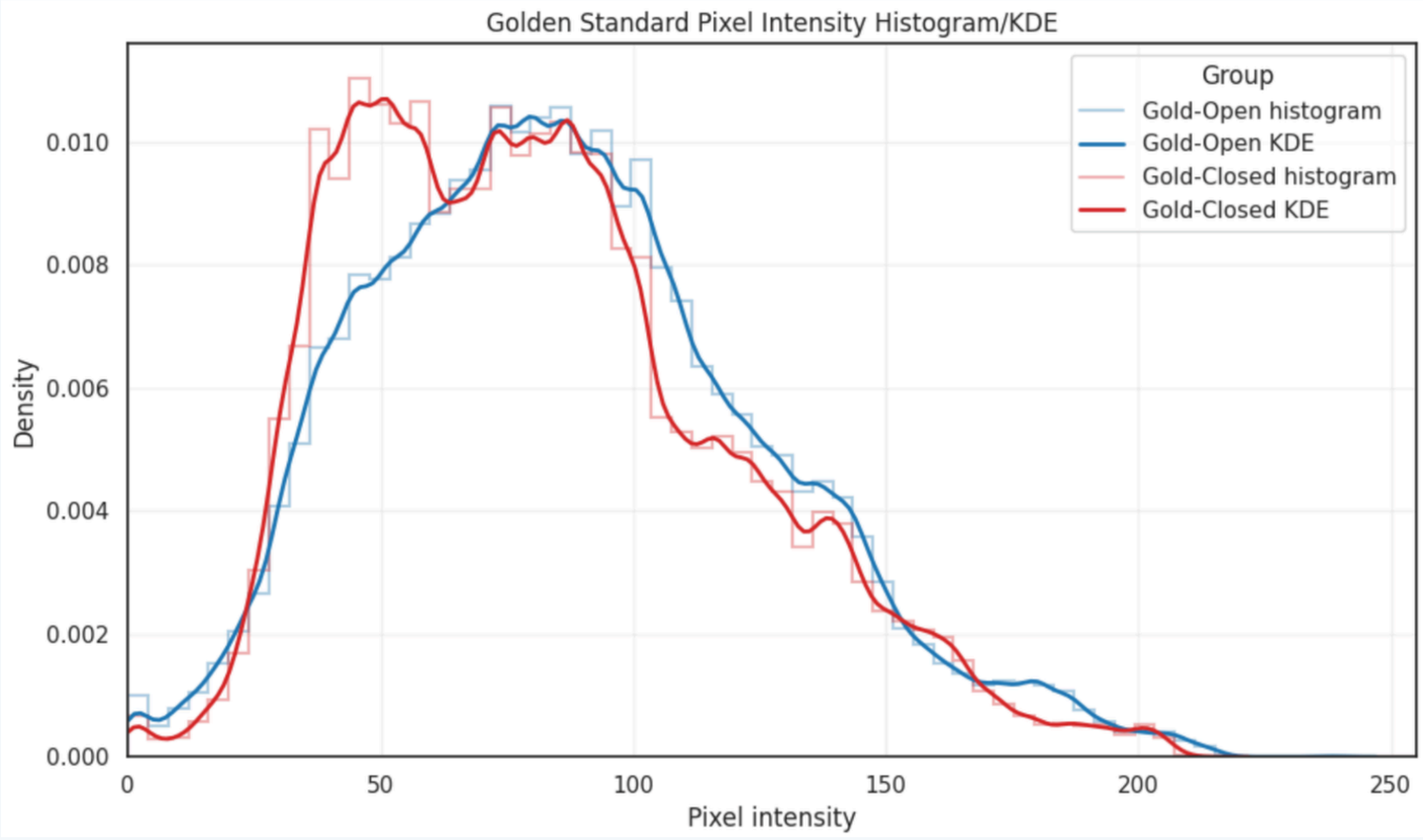
Open → Closed → Open
(EAR drops then rises = V-shape)

Image EDA | Average Image Analysis & Pixel Intensity Histogram

Image-level EDA is pending visual validation and must be completed before finalizing the feature pipeline.



Mean/Variance Comparison: Gold-Open vs Gold-Closed



Pixel Intensity Distribution: Gold-Open vs Gold-Closed

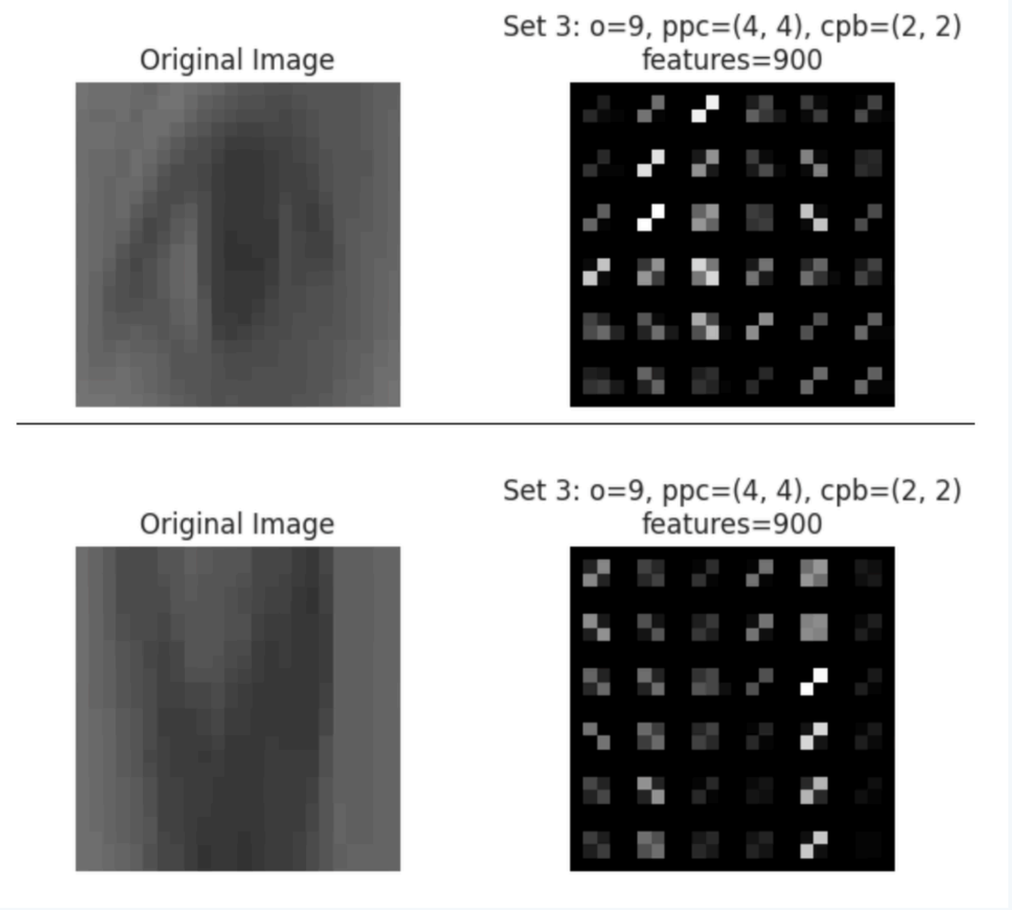
Image EDA | HOG Parameter Tuning

Problem:
The model needs a shared set of **HOG feature extraction parameters** for both classes, Open and Closed, combined with **brightness normalization** using **StandardScaler**. These parameters should be **robust to lighting changes** while still capturing the **curved shape of the pupil** and the **horizontal line of the eyelid**.

Evaluation Method:
Machine Learning Approach:

- **Linear SVM model** combined with **5-Fold Cross-Validation** was used directly on the **Golden Standard dataset of 1,500 images**.
- **Purpose**: measure how well each parameter set could separate the two classes.

Image EDA | HOG Parameter Tuning



HOG Feature Extraction Example

Set 1:

- 8 orientations, 4x4 pixels per cell, 1x1 cells per block — 288 features
- Accuracy of **81.87%**.
- The lack of **block normalization** made the features more sensitive to noise and lighting variations.

Set 2:

- 9 orientations, 8x8 pixels per cell, 2x2 cells per block — 144 features
- Achieving **87.47%** accuracy.
- It was fast, but the 8x8 cell size was too large for 24x24 eye images, causing the model to lose fine eyelid details.

Set 3:

- 9 orientations, 4x4 pixels per cell, 2x2 cells per block — 900 features
- Accuracy of **89.20%**.
- It provided the **best balance** between preserving fine eyelid details and normalizing shadow effects.

Key Findings & Modeling Implications

- ✔ Dataset is mostly clean — 0.49% data loss rate
- ⚠ Class imbalance must be handled — 77% open, 23% closed
- ✔ Member contribution is relatively balanced
- ✔ EAR varies across subjects — normalization required
- ✔ Blink should be modeled as a V-shape temporal sequence
- ✔ Image EDA shows that HOG + StandardScaler is needed to capture eye-state structure while reducing brightness and contributor bias.

Combine frame-level eye-state classification with sequence-level blink detection.